SkillLink

Service Marketplace System

Deliverable 2 · Sprint 2 Documentation

Module: Search & Discovery | Booking & Scheduling

Methodology: Scrum (Iterative & Incremental)

Team ENIGMA

Zaki Nabeel (Team Lead) · Shehryar Faisal (SCRUM Master) · Muhammad Sameer (Developer)

Roll Numbers: 23I-0604 · 23I-0508 · 23I-0634

CS3009 – Software Engineering | Spring 2026

Submission Date: 25th March, 2026

Part B — Sprint 2: Scrum Artifacts

B.1 Sprint 2 Overview

Sprint Number: Sprint 2

Sprint Duration: 2nd March 2026 – 22nd March 2026 (3 weeks)

Sprint Goal: Enable users to search and discover service providers, and allow clients to submit and manage booking requests.

Modules Developed: Search & Discovery | Booking & Scheduling

Leftover from Sprint 1: US-04: Client Contact Management (completed in this sprint)

B.2 Sprint 2 Backlog & User Stories

The following user stories were selected for Sprint 2, including the leftover item from Sprint 1. Each story follows the standard template: As a [type of user], I want [goal] so that [reason].

US-04 (Leftover): Client Contact Management

Story ID	US-04 — Leftover from Sprint 1
User Story	As a Client, I want to manage my contact information and location, so that service providers can accurately reach me to fulfil service requests.
Sub-Stories	4.1 Update phone number with regional validation (11-digit PK format). 4.2 Set physical address / location for geographic service filtering. 4.3 Allow clients to save multiple delivery addresses.
Given	Client is logged in and navigates to Profile Settings.
When	Client enters or updates a phone number or address and clicks Save.
Then	System validates the input, saves it to the database, and displays a success confirmation toast.

US-05: Service Search & Filtering

Story ID	US-05
User Story	As a Client, I want to search for service providers by skill category, location, and rating, so that I can quickly find the most suitable professional for my needs.
Sub-Stories	5.1 Implement a full-text search bar that queries provider names and skill tags. 5.2 Filter results by skill category (from taxonomy), city/area, and minimum star rating. 5.3 Display paginated search results with provider avatar, name, top skill, and rating.
Given	Client is logged in and navigates to the Search/Discover page.
When	Client types a keyword or selects filters and submits the search.
Then	System returns a ranked, paginated list of matching service providers within 2 seconds.

US-06: Provider Profile View

Story ID	US-06
User Story	As a Client, I want to view a service provider's full public profile, so that I can evaluate their skills, portfolio, ratings, and reviews before making a booking.
Sub-Stories	6.1 Display provider bio, skill categories, portfolio links, and profile photo. 6.2 Show aggregate rating and individual client reviews with timestamps. 6.3 Include a prominent 'Book Now' call-to-action button on the profile page.
Given	Client has executed a search and is viewing the results list.
When	Client clicks on a provider card to open the full profile.
Then	System renders the provider's complete public profile page with all details and the Book Now button enabled.

US-07: Booking Request Submission

Story ID	US-07
User Story	As a Client, I want to submit a booking request to a service provider with a preferred date/time and job description, so that I can formally engage them for a service.
Sub-Stories	7.1 Present a booking form with date picker, time picker, and job description text area. 7.2 Validate that the selected date is not in the past and the description is at least 20 characters. 7.3 Send an in-app notification to the provider upon booking submission.
Given	Client is viewing a provider's public profile and is authenticated.
When	Client fills in the booking form and clicks 'Confirm Booking'.
Then	System saves the booking record with status 'Pending', notifies the provider, and shows the client a booking confirmation screen.

US-08: Booking Management Dashboard

Story ID	US-08
User Story	As a Service Provider, I want to view, accept, or decline incoming booking requests, so that I can manage my schedule and commitments efficiently.
Sub-Stories	8.1 Display all incoming requests in a dashboard with status labels (Pending / Accepted / Declined). 8.2 Provider can accept a booking, which triggers a confirmation notification to the client. 8.3 Provider can decline a booking with an optional reason message sent to the client.
Given	Provider is logged in and navigates to the Bookings section of their dashboard.
When	Provider reviews a Pending booking card and clicks Accept or Decline.
Then	System updates the booking status in the database and sends the appropriate notification to the client in real time.

B.3 Structured Specifications

The following four structured specifications cover key user stories developed in Sprint 2. Each spec details the precise inputs, processing logic, outputs, pre/postconditions, and exception handling.

Spec-05.1 — Full-Text Provider Search

Spec ID	Spec-05.1
Description	Accept a user-supplied query string and return a ranked list of matching service providers.
Preconditions	User is authenticated as a Client. Search index is available and synced with the provider database.
Inputs	query: string (1–100 chars). Optional filters: categoryId (integer), city (string), minRating (float 1.0–5.0), page (integer, default 1).
Processing	1. Sanitise and trim the query string (strip special chars). 2. Execute full-text search across provider name and skill tag fields. 3. Apply category, city, and rating filters as WHERE clauses. 4. Rank results by rating DESC, then completeness of profile. 5. Paginate: return 10 results per page with total count.
Outputs	JSON array of provider objects: { providerId, name, avatar_url, topSkill, avgRating, city }. Pagination metadata: { page, totalPages, totalResults }.
Exceptions	EmptyQuery: return validation error 'Please enter a search term'. NoResults: return empty array with message 'No providers found — try broadening your search'. DBTimeout: return HTTP 503 with retry message.

Spec-06.1 — Public Provider Profile Rendering

Spec ID	Spec-06.1
Description	Fetch and render a complete provider profile page for a given provider ID.
Preconditions	Provider account exists and is active. Client is authenticated.
Inputs	providerId: integer (path parameter). viewerId: integer (from JWT session).
Processing	1. Validate providerId exists in DB; return 404 if not found. 2. Fetch provider record including bio, skills, portfolio URLs, and profile image URL. 3. Fetch aggregate rating: AVG(rating) from reviews table WHERE providerId. 4. Fetch latest 10 reviews ordered by createdAt DESC. 5. Log a profile view event for analytics. 6. Return assembled profile DTO to frontend.
Outputs	Provider profile page rendered with: avatar, name, bio, skill badges, portfolio icons, star rating (X.X / 5.0), review count, individual reviews, and 'Book Now' button.
Exceptions	ProviderNotFound (404): display 'This provider does not exist.' ProviderSuspended: show 'Account currently under review.' No reviews: display 'No reviews yet — be the first to book!'

Spec-07.1 — Booking Request Creation

Spec ID	Spec-07.1
Description	Validate and persist a new booking request submitted by a Client.
Preconditions	Client is authenticated. Provider profile is active. Client has completed their contact profile (US-04).
Inputs	providerId: integer. clientId: integer (from JWT). preferredDate: date string (YYYY-MM-DD). preferredTime: time string (HH:MM). jobDescription: string (20–1000 chars).
Processing	1. Validate preferredDate is not in the past; reject if so. 2. Validate jobDescription length (>= 20 chars). 3. Check provider is not already booked at that slot (conflict check). 4. Insert booking record: { clientId, providerId, preferredDate, preferredTime, jobDescription, status: 'Pending', createdAt: now() }. 5. Dispatch in-app notification to provider. 6. Return booking confirmation with bookingId.
Outputs	HTTP 201 with booking object: { bookingId, status: 'Pending', providerId, preferredDate, preferredTime }. In-app notification delivered to provider.
Exceptions	PastDate: 'Selected date has already passed.' DescriptionTooShort: 'Please provide at least 20 characters describing your job.' SlotConflict: 'Provider is unavailable at this time — please choose another slot.' ProviderInactive: 'This provider is currently unavailable.'

Spec-08.1 — Booking Accept / Decline by Provider

Spec ID	Spec-08.1
Description	Allow a Service Provider to accept or decline a pending booking request.
Preconditions	Provider is authenticated. Booking exists with status 'Pending'. Booking belongs to this provider.
Inputs	bookingId: integer. action: enum { 'accept', 'decline' }. declineReason: string (optional, max 200 chars, required only when action = 'decline').
Processing	1. Fetch booking by bookingId; verify it belongs to the authenticated provider. 2. Verify current status is 'Pending'; reject if already actioned. 3. If action = 'accept': update status to 'Accepted', send confirmation notification to client. 4. If action = 'decline': update status to 'Declined', attach declineReason, send decline notification to client. 5. Log action timestamp.
Outputs	HTTP 200 with updated booking object: { bookingId, status: 'Accepted' | 'Declined', updatedAt }. Push notification delivered to client.
Exceptions	BookingNotFound (404): 'Booking does not exist.' UnauthorisedProvider: 'You are not authorised to action this booking.' AlreadyActioned: 'This booking has already been accepted or declined.' DeclineReasonMissing: 'Please provide a reason when declining a booking.'

B.4 Scrum Board — Trello Snapshots

The Sprint 2 Trello board contains five columns: Product Backlog | Sprint Backlog | In Progress | Done | Leftover List. Snapshots were taken at three key milestones.

Snapshot 1 — Sprint Planning (Start of Sprint 2)

All five user stories (US-04 leftover + US-05, US-06, US-07, US-08) loaded into the Sprint Backlog column. In Progress and Done columns are empty. The sprint goal card is pinned at the top of the Sprint Backlog.

[Trello Board Screenshot — Sprint Planning Stage]
Sprint Backlog: US-04, US-05, US-06, US-07, US-08 | In Progress: — | Done: —

Snapshot 2 — Mid-Sprint Progress

US-04 and US-05 moved to Done. US-06 and US-07 are actively In Progress. US-08 remains in the Sprint Backlog awaiting its turn.

[Trello Board Screenshot — Mid-Sprint Stage]
Done: US-04, US-05 | In Progress: US-06, US-07 | Backlog: US-08

Snapshot 3 — Sprint Completion

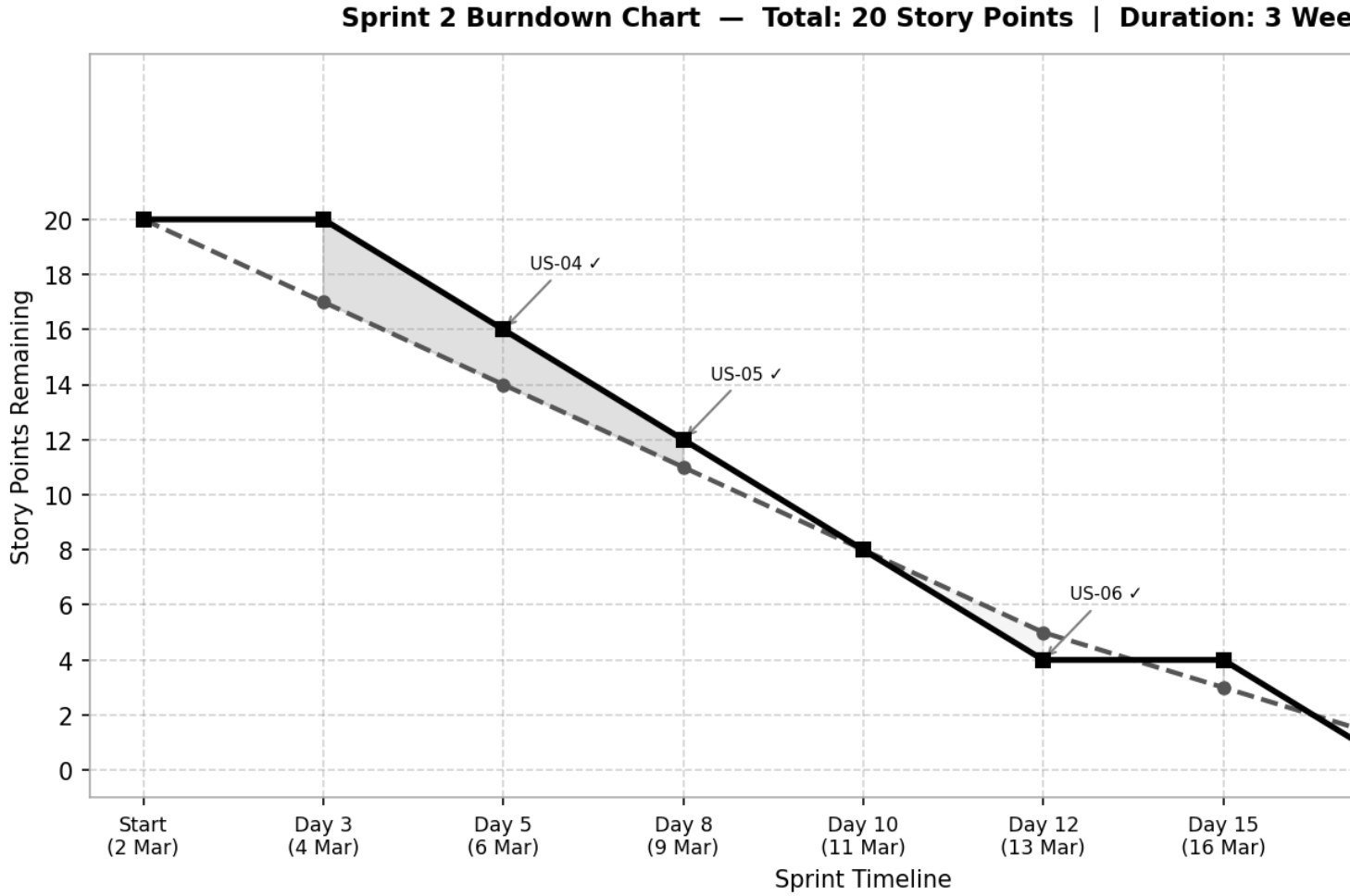
US-04, US-05, US-06, US-07, and US-08 all moved to Done. No leftover stories for Sprint 3. Sprint retrospective notes are attached to the Sprint Goal card.

[Trello Board Screenshot — Sprint Completion Stage]
Done: US-04, US-05, US-06, US-07, US-08 | In Progress: — | Leftover: —

User Story	Sprint	Final Status	Notes
US-04: Client Contact Mgmt	Sprint 2 (Leftover)	**Done**	Completed — carried from Sprint 1
US-05: Service Search & Filtering	Sprint 2	**Done**	Completed within Sprint 2
US-06: Provider Profile View	Sprint 2	**Done**	Completed within Sprint 2
US-07: Booking Request Submission	Sprint 2	**Done**	Completed within Sprint 2
US-08: Booking Mgmt Dashboard	Sprint 2	**Done**	Completed within Sprint 2

B.5 Sprint 2 Burn-Down Chart

The burn-down chart below tracks planned story points versus actual completed story points across the 3-week Sprint 2 duration. Each user story was assigned 4 story points, giving a total sprint capacity of 20 points.



Date / Week	Ideal Remaining	Actual Remaining	Points Done	Story Completed
Sprint Start (2 Mar)	20 pts	20 pts	0 pts	—
Day 3 (4 Mar)	17 pts	20 pts	0 pts	—
Day 5 (6 Mar)	14 pts	16 pts	4 pts	US-04 ✓
Day 8 (9 Mar)	11 pts	12 pts	8 pts	US-05 ✓
Day 10 (11 Mar)	8 pts	8 pts	12 pts	—
Day 12 (13 Mar)	5 pts	4 pts	16 pts	US-06 ✓
Day 15 (16 Mar)	3 pts	4 pts	16 pts	—

Day 17 (18 Mar)	1 pts	0 pts	20 pts	US-07 ✓ US-08 ✓
Sprint End (22 Mar)	0 pts	0 pts	20 pts	All stories Done

Sprint Outcome: All 5 planned story points (20 pts total) were completed. The team experienced a minor 2-day delay at the start due to backlog refinement, but recovered ahead of schedule by Day 17. No leftover stories for Sprint 3.

Part C — Deliverable Document

C.1 Introduction

SkillLink is a digital service marketplace platform connecting clients who need skilled professionals with verified service providers across trades and expertise areas in Pakistan. The platform addresses a gap in the local market: the absence of a trustworthy, centralised portal where clients can discover, vet, and seamlessly book providers such as plumbers, electricians, graphic designers, tutors, and software developers.

Sprint 1 established the foundational architecture for user onboarding and role-based profile management, delivering three of four planned user stories (US-01: Account Registration, US-02: Secure Authentication, US-03: Provider Profile Setup). US-04 (Client Contact Management) was carried forward as a leftover.

Sprint 2 builds directly on this foundation. It delivers the leftover US-04 alongside four new user stories (US-05 through US-08), introducing the Search & Discovery and Booking & Scheduling modules. Together, these modules form the core transactional layer of the SkillLink platform — enabling clients to find providers and initiate service engagements.

C.2 Project Vision

The project vision established in Deliverable 1 remains unchanged and continues to guide all development decisions:

> *To become the leading digital service marketplace in Pakistan, empowering clients to access quality services instantly while enabling skilled providers to grow their businesses through a transparent, secure, and user-friendly platform.*

Sprint 2 directly advances this vision by making the platform's search, discovery, and booking capabilities operational — transforming SkillLink from a registration system into a functioning marketplace.

C.3 Intended Use of the System — Stakeholders

No changes were made to the stakeholder landscape during Sprint 2. The following table reflects updated needs based on Sprint 2 feature delivery:

Stakeholder	Role	Updated Needs (Sprint 2)
Clients	End users seeking services	Search providers by skill/location, view full profiles, submit booking requests, manage contact info

Service Providers	Professionals offering skills	Receive and action booking requests, profile visibility in search results, manage booking history
Platform Admin	System moderator	Monitor booking activity, resolve disputes, review flagged providers
Dev Team	Builders & maintainers	Clear search and booking specs, testable acceptance criteria, stable API contracts

C.4 Features & Functionality

Sprint 1 Features (Completed)

- Account Registration — role-based signup (Client / Provider) with email validation and strong password policy
- Secure Authentication — JWT-based login, Forgot Password flow, Remember Me session persistence
- Provider Profile Setup — profile photo upload, skill category selection, professional bio, portfolio link integration

Sprint 2 Features (New)

- Client Contact Management (US-04) — phone number with regional PK validation, address management, multiple saved locations
- Service Search & Filtering (US-05) — full-text search across provider names and skill tags; filters by category, city, and rating; paginated results
- Provider Profile View (US-06) — full public profile rendering with bio, skills, portfolio, aggregate rating, and individual client reviews
- Booking Request Submission (US-07) — booking form with date/time picker and job description; real-time slot conflict detection; provider notification on submission
- Booking Management Dashboard (US-08) — provider dashboard to view, accept, or decline incoming requests with automated client notifications

Sprint 2 Enhancement Summary

Sprint 2 elevated SkillLink from a static profile system to a live, transactional marketplace. Clients can now discover providers through a structured search interface and formally engage them through a managed booking flow — a significant expansion of system capability. Provider-side workflow tooling (booking dashboard) closes the communication loop between the two primary user roles.

C.5 User Stories — Sprint 2

The following user stories were addressed in Sprint 2. Detailed backlog tasks and acceptance criteria are documented in Part B Section 2.

ID	User Story	As a [user], I want [goal] so that [reason]	Status
US-04	Client Contact Mgmt (Leftover)	As a Client, I want to manage my contact information and location, so that providers can accurately reach me.	**Done ✓**
US-05	Service Search & Filtering	As a Client, I want to search for providers by skill, location, and rating, so that I can find the right professional quickly.	**Done ✓**
US-06	Provider Profile View	As a Client, I want to view a provider's full profile, so that I can evaluate their skills and reviews before booking.	**Done ✓**
US-07	Booking Request Submission	As a Client, I want to submit a booking request with date/time and description, so that I can formally engage a provider.	**Done ✓**
US-08	Booking Mgmt Dashboard	As a Provider, I want to view, accept, or decline bookings, so that I can manage my schedule efficiently.	**Done ✓**

C.6 Non-Functional Requirements (NFR) — Updated

The NFRs defined in Deliverable 1 have been reviewed and refined for Sprint 2. New requirements addressing search performance and booking reliability have been added.

#	Category	NFR Statement	Priority
1	Performance	The system shall load any page within 3 seconds for up to 500 concurrent users.	High
2	Performance	The search API shall return results within 2 seconds for queries against a dataset of 10,000+ provider records.	High
3	Security	All passwords shall be hashed using bcrypt; no plain-text credentials shall be stored at rest.	Critical
4	Security	All API communication shall use HTTPS/TLS 1.2 or higher to prevent interception.	Critical
5	Security	Booking actions (accept/decline) shall require the authenticated provider's JWT; unauthorised requests shall return HTTP 403.	Critical
6	Availability	The platform shall maintain 99.5% uptime; planned maintenance shall be communicated 24 hours in advance.	High
7	Scalability	The search and booking modules shall support horizontal scaling to handle a 10x increase in concurrent users without degradation.	Medium
8	Usability	The interface shall conform to WCAG 2.1 AA accessibility standards; search results must be keyboard-navigable.	High
9	Reliability	All booking state transitions shall be atomic; partial failures shall trigger a rollback and surface a user-readable error.	High
10	Reliability	User data shall be auto-backed up every 24 hours with a 4-hour restore capability.	High
11	Maintainability	Source code shall follow agreed coding standards and be documented for 2-day developer onboarding.	Medium

C.7 Iteration 2 — Implementation Screenshots

The following screenshots document the new modules, UI enhancements, and system functionality delivered in Sprint 2. Each screenshot includes a caption describing its functionality.

Screenshot 1 — Contact Management Form (US-04)

[Screenshot Placeholder — attach actual screenshot here]

Caption: The client profile settings screen displaying phone number input with real-time 11-digit PK format validation and address fields with location saving functionality.

Screenshot 2 — Search Page with Filters (US-05)

[Screenshot Placeholder — attach actual screenshot here]

Caption: The main search interface showing the full-text search bar, category dropdown filter, city selector, and minimum rating slider. Results appear as provider cards with avatar, name, top skill, and star rating.

Screenshot 3 — Search Results Grid (US-05)

[Screenshot Placeholder — attach actual screenshot here]

Caption: Paginated search results displayed in a responsive card grid. Each card shows provider avatar, verified badge, top skill category, city, and aggregate star rating. Pagination controls appear at the bottom.

Screenshot 4 — Provider Public Profile Page (US-06)

[Screenshot Placeholder — attach actual screenshot here]

Caption: The full provider profile page showing profile photo, professional bio, skill badge list, portfolio link icons, aggregate rating (e.g., 4.7 / 5.0 from 23 reviews), individual client reviews with timestamps, and the prominent 'Book Now' button.

Screenshot 5 — Booking Request Form (US-07)

[Screenshot Placeholder — attach actual screenshot here]

Caption: The booking modal triggered from the provider profile. Shows a date picker (past dates greyed out), time picker, job description text area with live character counter, and a 'Confirm Booking' button that activates once all fields are valid.

Screenshot 6 — Booking Confirmation Screen (US-07)

[Screenshot Placeholder — attach actual screenshot here]

Caption: Post-submission confirmation screen showing booking ID, provider name, selected date/time, and a status badge reading 'Pending'. A notification icon confirms the provider has been alerted.

Screenshot 7 — Provider Booking Dashboard (US-08)

[Screenshot Placeholder — attach actual screenshot here]

Caption: The provider's booking management dashboard displaying incoming requests as cards with status labels (Pending / Accepted / Declined). Each card contains client name, job description, preferred date, and Accept/Decline action buttons.

Screenshot 8 — Booking Action Modal (US-08)

[Screenshot Placeholder — attach actual screenshot here]

Caption: The decline booking modal where the provider enters an optional reason for declining. The 'Send & Decline' button dispatches a notification to the client with the provided reason.

C.8 Work Division — Sprint 2

Sprint 2 responsibilities were distributed based on each member's core competencies, consistent with the work division pattern established in Sprint 1.

Team Member	Role	Sprint 2 Responsibilities
Zaki Nabeel 23l-0604 (Team Lead)	Project Manager · Tester	Sprint 2 planning and milestone tracking. Defined acceptance criteria for US-05, US-06, US-07, US-08. Test case design and QA validation for search and booking modules. Burn-down chart maintenance. Final deliverable packaging and submission.
Shehryar Faisal 23l-0508 (SCRUM Master)	SCRUM Master · Analyst / Architect	Sprint facilitation: daily standups, backlog grooming, sprint review. Authored all four structured specifications (Spec-05.1, Spec-06.1, Spec-07.1, Spec-08.1). System architecture design for search indexing and booking state machine. Trello board management and snapshot capture.
Muhammad Sameer 23l-0634 (Developer)	Developer · UI Designer	Front-end implementation of search page (US-05), provider profile page (US-06), booking request form (US-07), and booking management dashboard (US-08). React component development, responsive layout, UI/UX design for all Sprint 2 screens. Completed US-04 contact management front-end.

Sprint 2 Outcome: All 5 user stories (20 story points) were completed within the sprint. The SkillLink platform now supports end-to-end service discovery and booking initiation. Sprint 3 will focus on Reviews & Ratings, In-App Messaging, and Payment Integration.